Introduction & History

What is Image Processing?

Digital Image Processing consists of the various techniques and methods involving in the manipulation of images on a computer. Various types of operations are performed on images, which constitute Digital Image Processing.

What is Comptuer Vision?

Computer vision is a field of artificial intelligence (AI) that enables computers and systems to derive meaningful information from digital images, videos, and other visual inputs — and take actions or make recommendations based on that information. If AI enables computers to think, computer vision enables them to see, observe and understand.

A Brief HISTORY

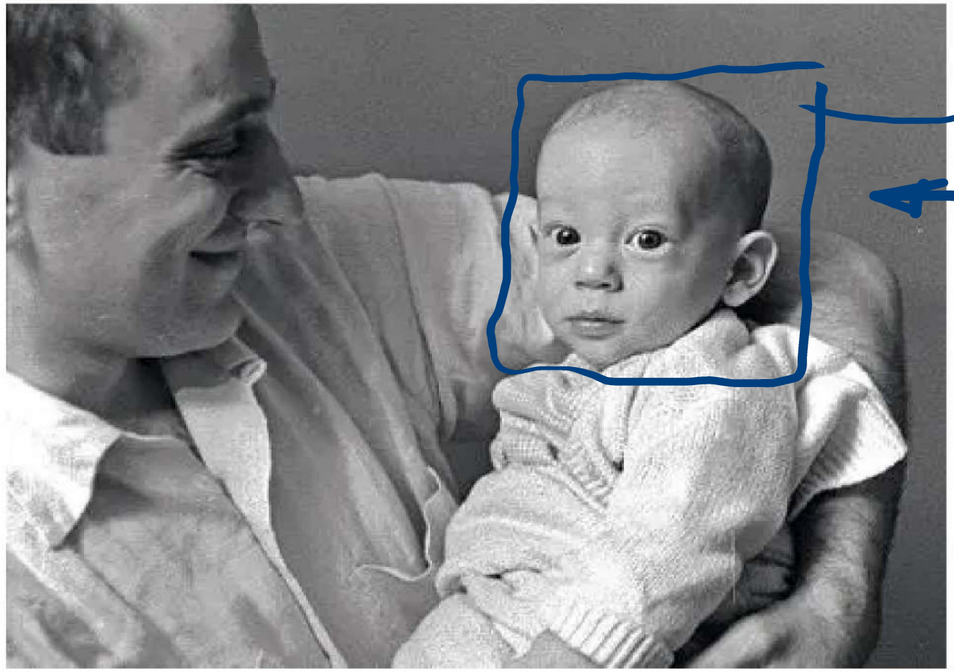
1920s



A Brief HISTORY

1950s





A Brief HISTORY

1960s



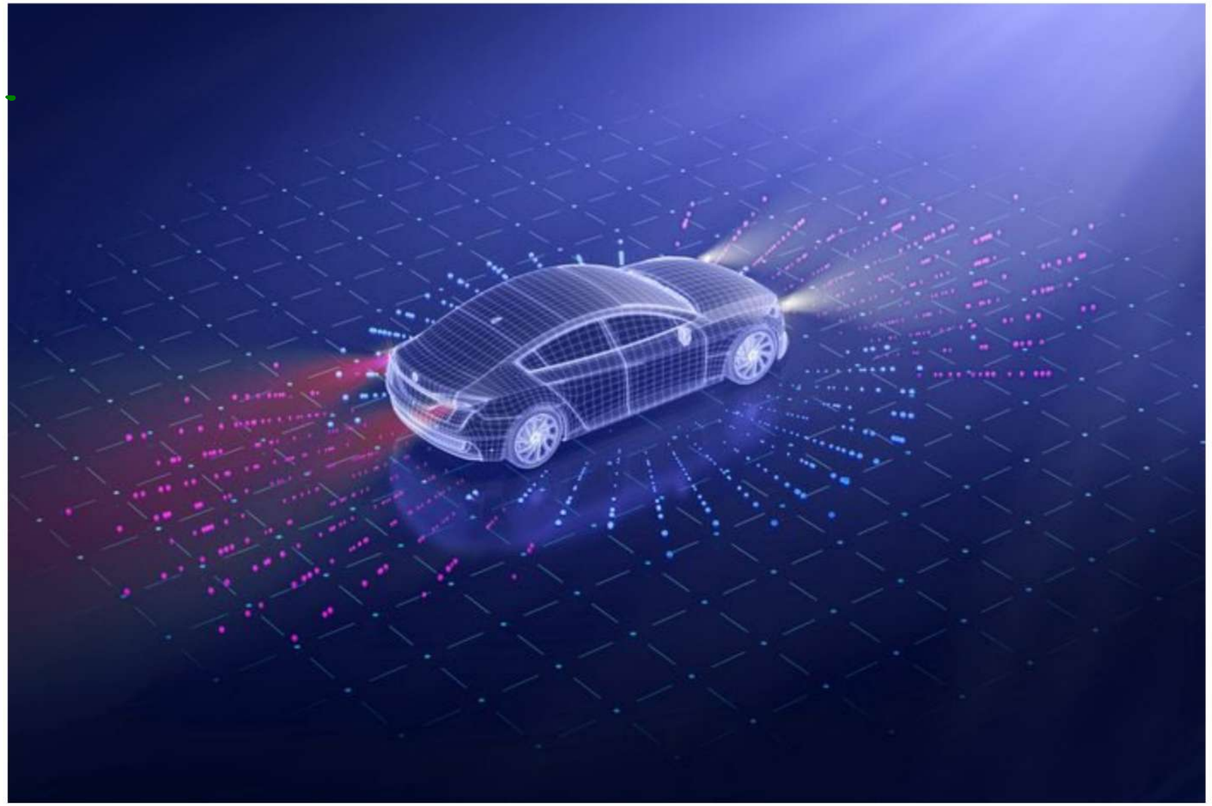
A Brief HISTORY

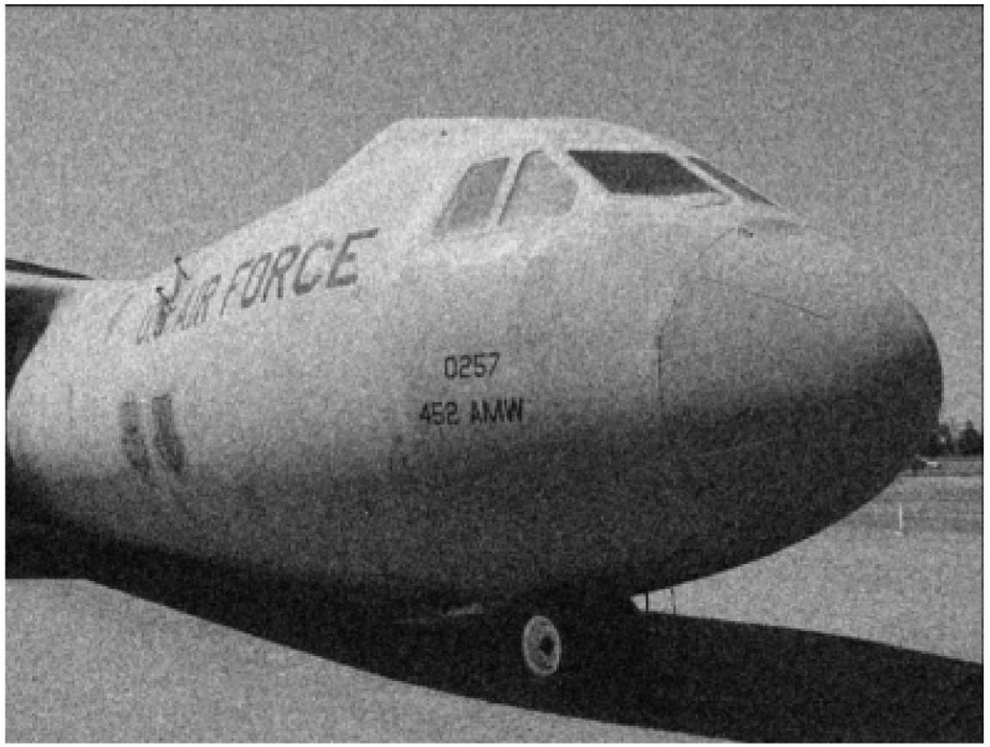
1970s



A Brief HISTORY

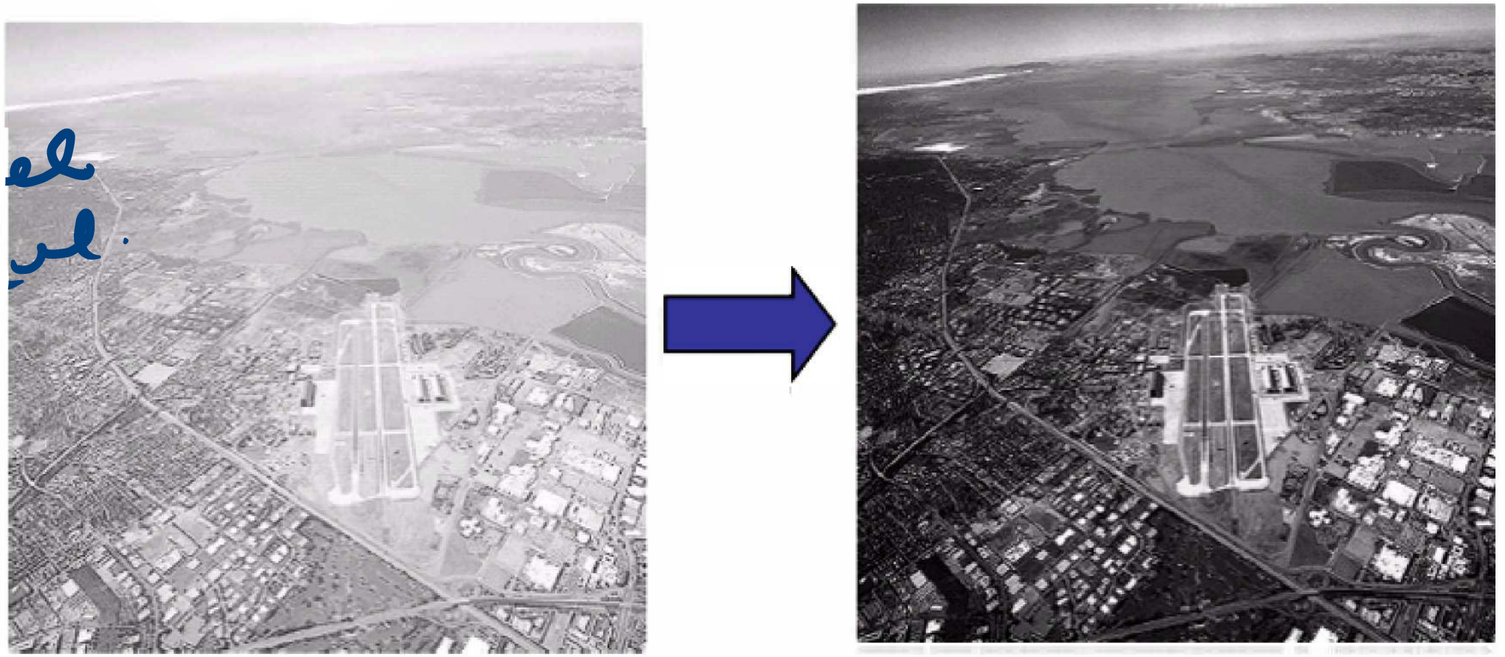
1980s - Todays



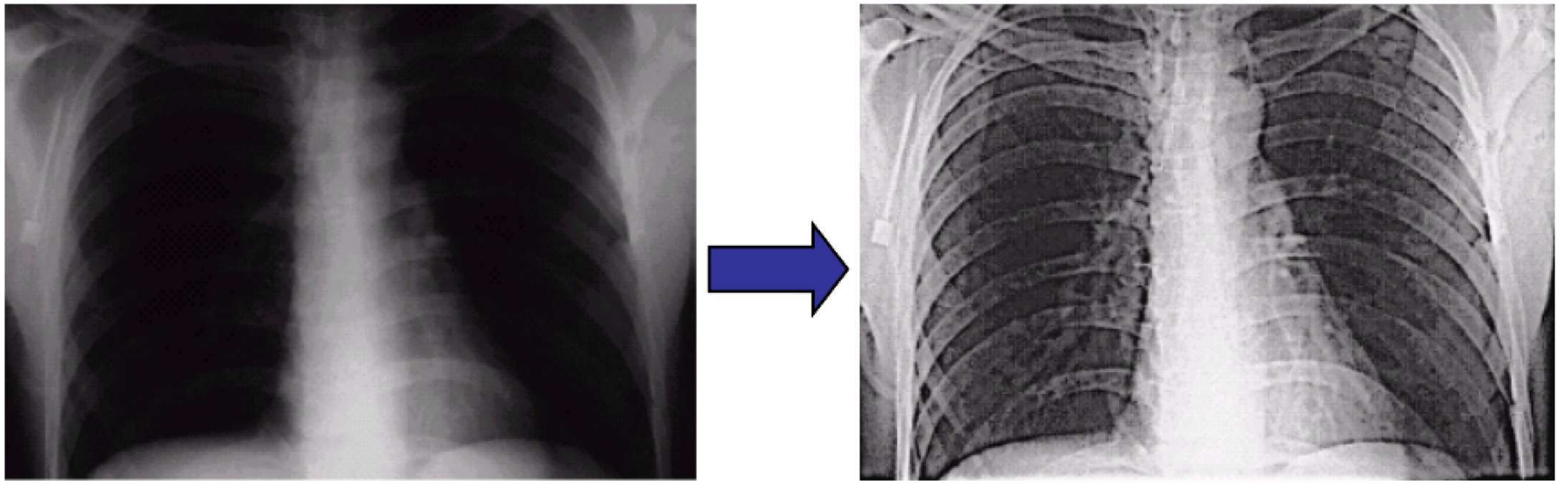




Noise Removal



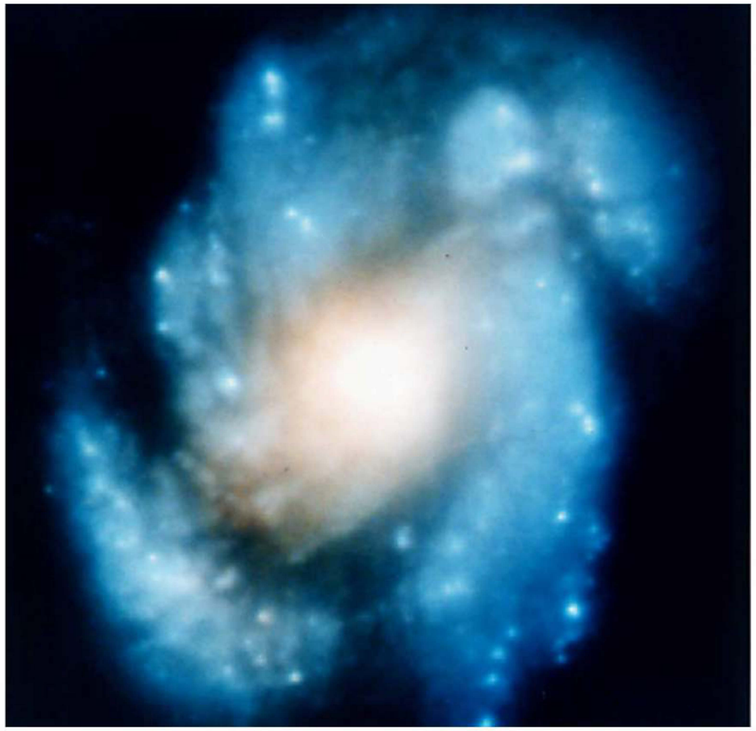
Contrast Adjustment



Contrast Adjustment



Launched in 1990 the Hubble telescope can take images of very distant objects However, an incorrect mirror made many of Hubble's images useless Image processing techniques were used to fix this.



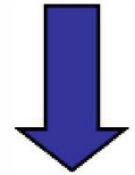




Image Enhancement





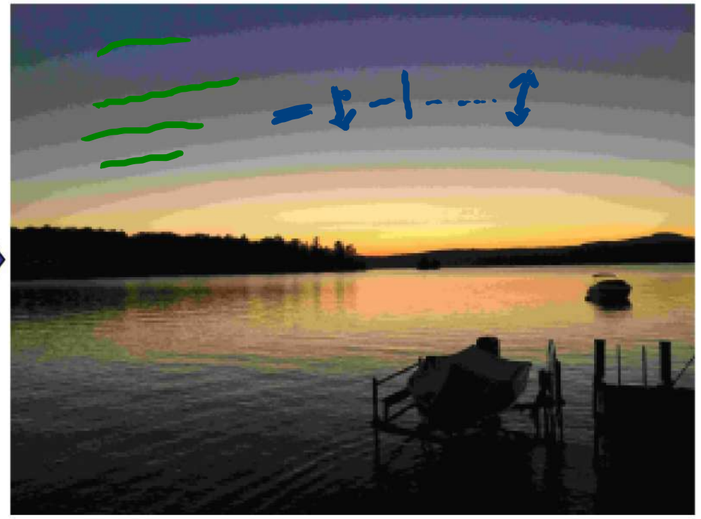
Edge Detection



Original (1.9M)



Compressed (230 KB. 12%)



Compressed (120 KB, 7%)

Compression



Damaged



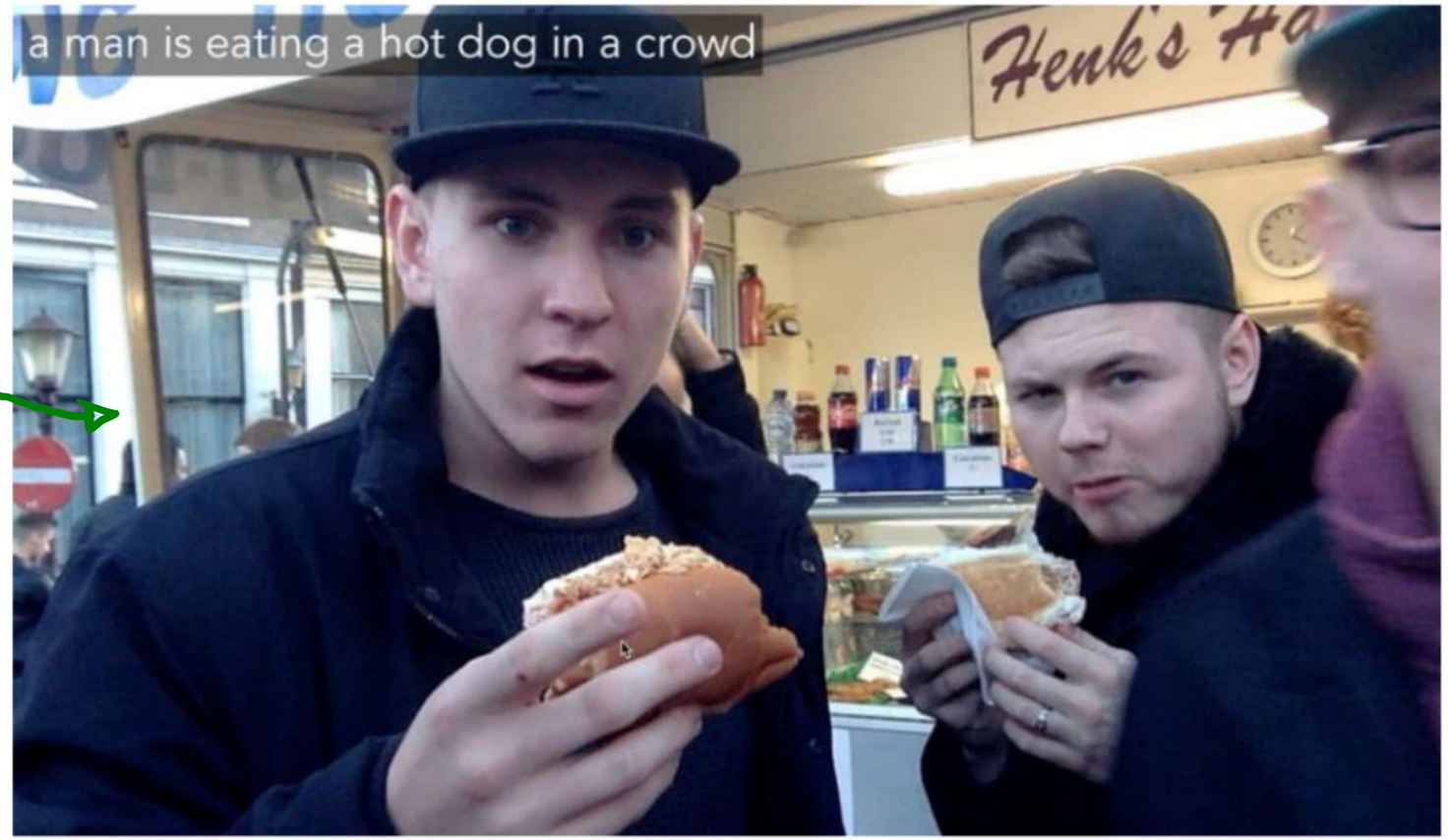
Restored

Restoration



- Can you find the aircraft?

Object Detection



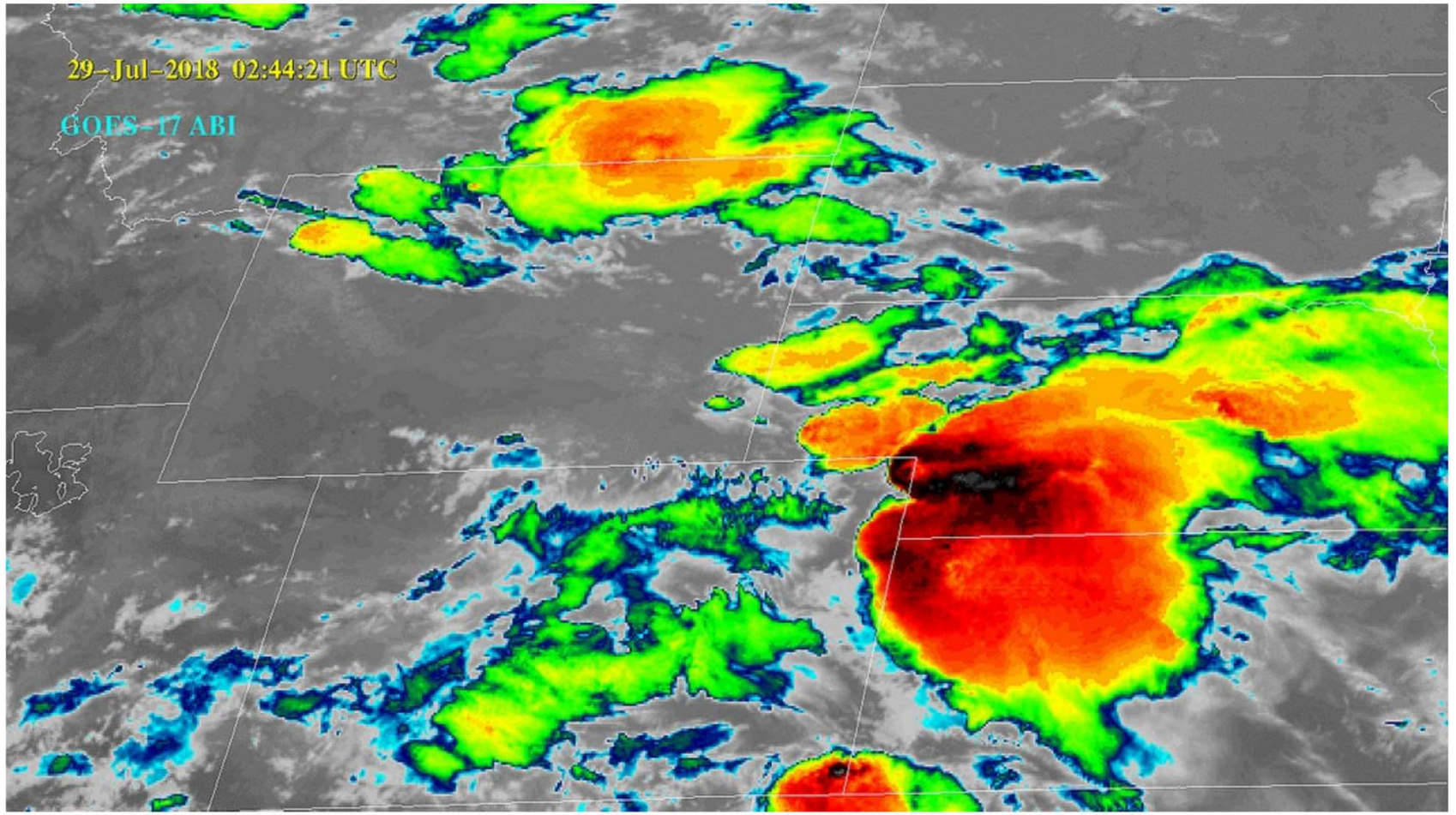
Scene Understanding

- 2004 - 142 miles of desert road, 15 entries
- None finished
- Today cars can safely self drive on urban roads!

- 2007 Entry (Link)



UseCase : DARPA Grand Challenge



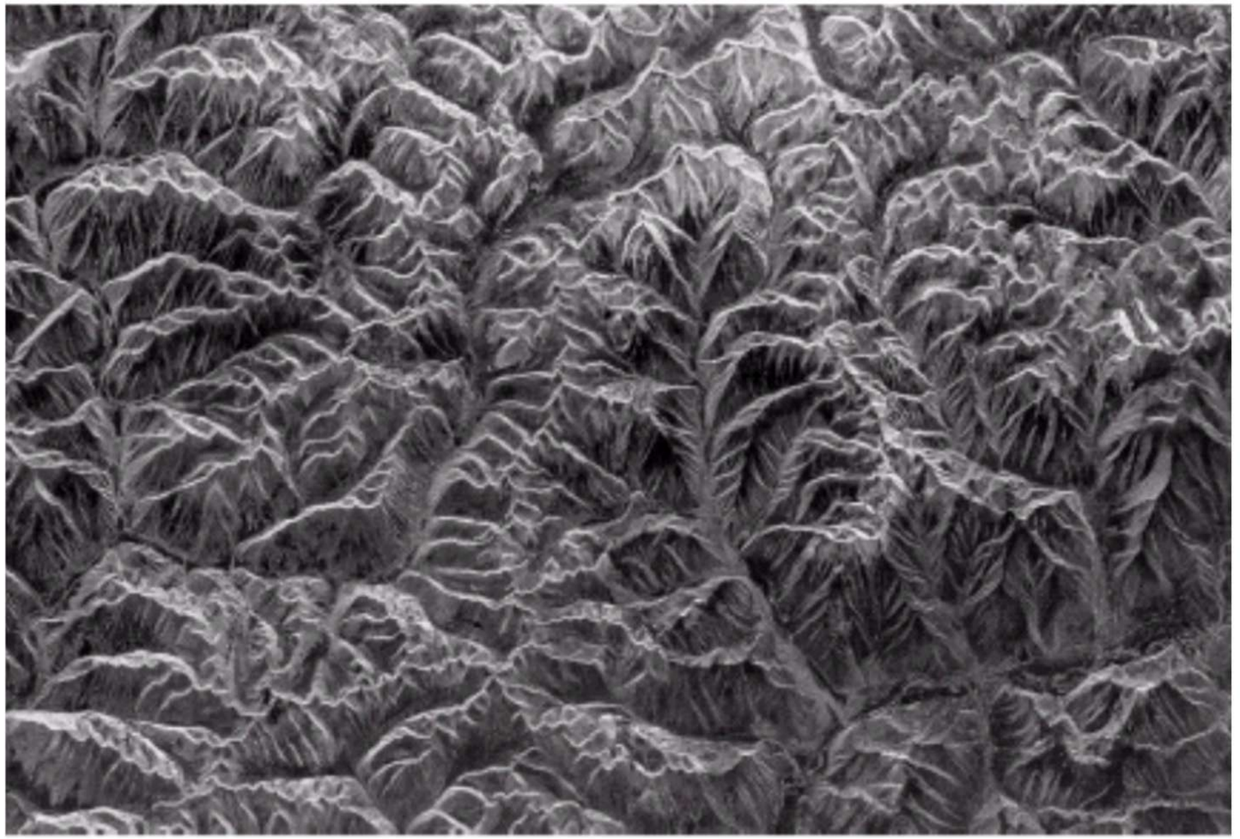
Infrared Satellite Imaging



a b
c
d

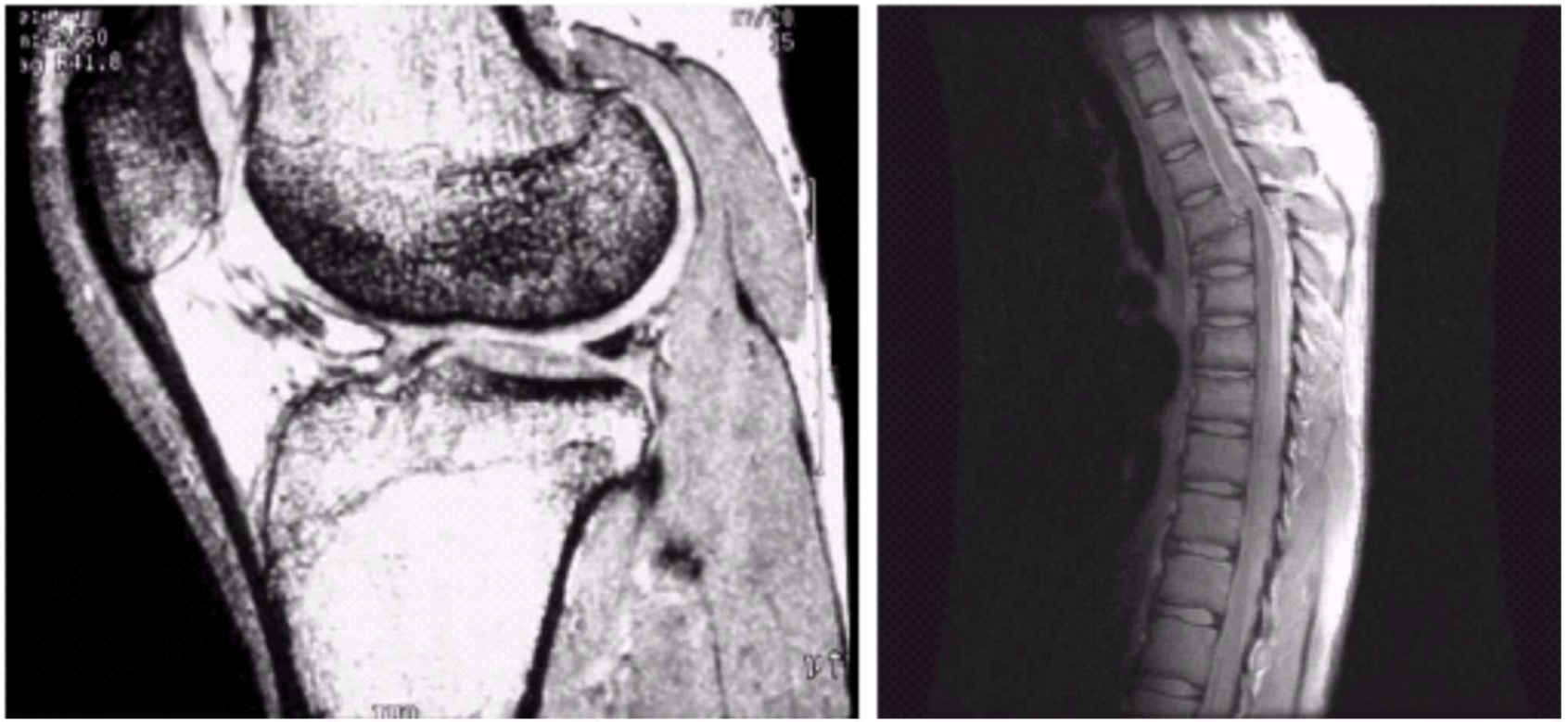
Some additional examples of imaging in the visual spectrum. (a) Thumb print. (b) Paper currency. (c) and (d) Automated license plate reading. (Figure (a) courtesy of the National Institute of Standards and Technology. Figures (c) and (d) courtesy of Dr. Juan Herrera, Perceptics Corporation.)

Automated Visual Inspection(2)



Spaceborne radar image of mountains in southeast Tibet. (Courtesy of NASA.)

Example of Radar Image



a b

MRI images of a human (a) knee, and (b) spine. (Image (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

MRI (Radio Band)